CertiVar: Certified Likelihood Ratios for Variant-Effect Predictor Calls, Calibrated on Saturation Mutagenesis Data

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Abstract

Variant-effect predictors are a common source of evidence for missense variants of uncertain significance (VUS), and the Bayesian form of the ACMG/AMP clinical guidelines expresses evidence strength as a likelihood ratio. Gene-specific calibrations estimate this ratio from few labelled variants, as point estimates or resampling-based bounds that are not designed to remain valid once a threshold has been chosen from the same data. We present CertiVar, which turns a predictor score into loss-of-function (LoF) calls, functional calls or abstentions with two class-conditional conformal thresholds learned from saturation genome editing or deep mutational scanning data, keeping one variant per residue because substitutions at a residue are dependent. We prove that exact bounds on the likelihood ratio of each call type with respect to the assay outcome remain valid after a threshold is selected from a grid with a Bonferroni correction, that fixed-sequence testing certifies evidence levels after selection, and that per-class coverage after selection is at least one minus the largest grid level. Because every guarantee conditions on the assay class, none requires the share of LoF variants among the variants being called, provided their within-class score distribution matches that of the assayed variants; a mixture-consistency check tests a necessary condition for this on unlabelled targets. On 14,477 assayed missense variants in BRCA1, BRCA2, BAP1, RAD51C and MSH2, scored by AlphaMissense, REVEL and ESM-2, the lower of the two per-class coverages was 0.896–0.908 at a 0.90 target in all 15 gene–predictor pairs, with 35% abstention on average, and stayed at 0.89 or above when the calibration LoF share was varied from 5% to 50%; marginal conformal prediction calibrated at 5% covered LoF variants at only 0.11 in its worst pair. Against known population likelihood ratios, the certified bounds all held together in 99.7% of 1,500 resampled data sets (guaranteed: at least 90%), against 57.6% for percentile-bootstrap bounds. At the 5% level, the mixture check rejected 4.7% of random splits and every tested set of unassayed VUS outside the assayed residues. The certificates concern the assay, not pathogenicity; certified evidence was mostly supporting-level, required same-gene assay data, and held for 88.8% of bounds in held-out protein regions, below the nominal 95%. Code: `https://github.com/aaryavvatts-lab/certivar`

Keywords: variant effect prediction; conformal prediction; likelihood ratio; ACMG/AMP evidence calibration; multiplexed assays of variant effect; variants of uncertain significance; selective classification; multiple testing.

1 Introduction

In the five cancer-predisposition genes studied here, 81% of the 11,010 ClinVar missense records classified as (likely) pathogenic, (likely) benign or of uncertain significance are variants of uncertain significance (VUS; our count from the release of 26 October 2026 [1]), and for many of them computational predictors are among the few sources of evidence [2]. The ACMG/AMP guidelines of the American College of Medical Genetics and Genomics and the Association for Molecular Pathology grade evidence qualitatively [3]; their Bayesian adaptation equates each strength with a likelihood ratio, the odds of pathogenicity (OddsPath) [4, 5]. Calibration workflows report this ratio as a point estimate or a resampling-based bound [2, 6, 7], but for a single gene few labelled variants are available and the threshold is chosen from the same data; these bounds are not designed to remain valid after that choice. Two features of the data make the estimate fragile. The share of damaging variants differs between calibration sets and the variants being classified: 45% of confidently labelled ClinVar missense variants in our genes are pathogenic, whereas 6.0% to 25.9% of the other assayed variants lose function in saturation genome editing (SGE) or deep mutational scanning (DMS) experiments (Table 1). And labelled sets are small, with dependent substitutions at each residue: when the

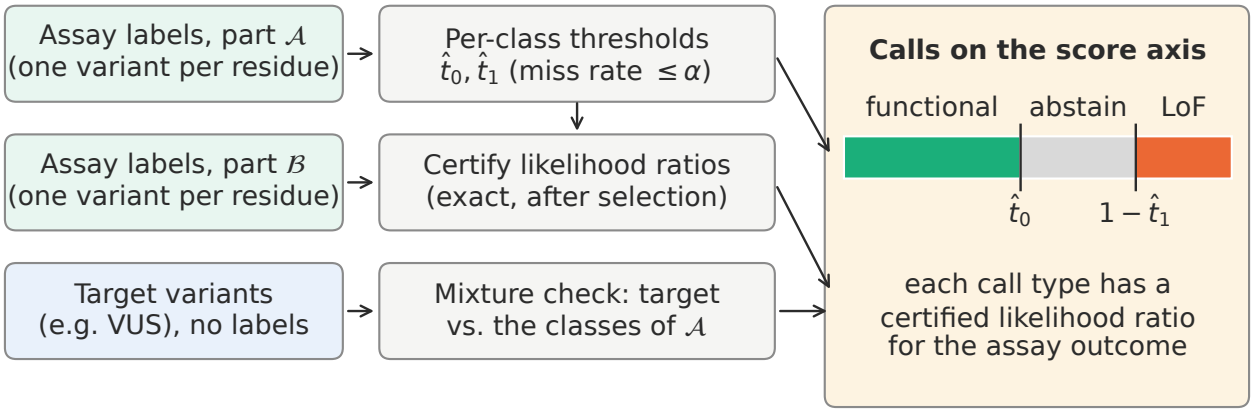


Figure 1: CertiVar. Per-class thresholds are set on part $\mathcal{A}$ of a gene's assayed residues, the likelihood ratio of each call type with respect to the assay outcome is certified on a disjoint part $\mathcal{B}$, and a mixture check asks whether the unlabelled target scores are a re-weighting of the two classes of $\mathcal{A}$. On the score axis the two thresholds give three outcomes.

population likelihood ratio was known, thresholds selected on percentile-bootstrap bounds respected every claimed bound in only 57.6% of resampled data sets (Section 4.4).

We introduce CertiVar (Figure 1), which turns a predictor score into loss-of-function (LoF), functional or no calls by two cut-offs: per-class (Mondrian) conformal thresholds, set from SGE or DMS labels with one variant per residue, bound how often each class is mis-called, and a certification step bounds the likelihood ratio of each call type with respect to the assay outcome after a threshold is selected from a grid. Neither step needs the LoF share of the variants being called, provided their within-class score distribution matches that of the assayed variants, and a mixture-consistency check tests a necessary condition for this. The certified quantity is evidence about pathogenicity only to the extent that the assay is clinically validated. Our contributions are: (1) certified likelihood ratios for both call types of an abstaining classifier on residue-clustered data, valid after threshold selection with a Bonferroni correction (evidence levels also with fixed-sequence testing), with a coverage guarantee after selection and disjoint call sets (Section 3); (2) a label-free check of class-conditional invariance, which kept close to its 5% level on random splits and rejected every tested set of unassayed VUS outside the assayed residues (Section 4.5); (3) a benchmark against known population likelihood ratios in which exact and Katz bounds held while bootstrap bounds and point estimates overstated evidence (Section 4.4); and (4) evidence on five genes and three predictors that marginal and prior-reweighted conformal prediction lose coverage of one class under a shift of the calibration LoF share, while per-class coverage holds (Section 4.2).

2 Related work

ClinGen maps predictor scores to ACMG/AMP evidence strengths through local likelihood ratios estimated against ClinVar labels with a one-sided bootstrap bound [2], extended to more predictors [6], to assay scores [5, 8, 9] and to gene- and domain-specific calibration [7, 10, 11], the last with sign-test bounds on resampled likelihood ratios [7]. Split conformal prediction gives finite-sample coverage [12]; its class-conditional (Mondrian) form controls coverage within each class [12–14]; order-statistic thresholds give training-conditional control [15–17]; keeping one member per group handles clustered data [18]; and abstaining classifiers trade coverage against error [19]. Selecting a threshold with exact binomial bounds and a union bound underlies selective classification with guaranteed risk [20] and risk-controlling prediction sets [21]; fixed-sequence testing over an ordered grid is the Learn-then-Test template [22]. Harrizi et al. reported split-conformal prediction sets for missense predictors on ClinVar at 90% coverage [23]; CertiVar instead controls coverage within each class, calibrates on assay labels and certifies the likelihood ratio of the calls. Large-sample intervals for a ratio of binomial proportions [24, 25] carry no finite-sample guarantee (Section 4.4 compares Katz intervals). Label-shift methods adjust posteriors or conformal weights to an estimated target prior [26–28]; class-conditional quantities need none, provided class-conditional invariance holds, which Section 3 tests.

3 Method

Setting. Each missense variant has a predictor score s, oriented so that higher means more damaging, and an assay label $Y \in \{0, 1\}$, with $Y = 1$ for loss of function (LoF). A *residue* is a protein position of one gene; the substitutions at a residue share structural context and form one cluster. A gene's assayed missense variants are split by residue into a calibration part $\mathcal{A}$ and a certification part $\mathcal{B}$; the variants to be called (for example VUS) form the target. We write $q(s) \in [0, 1]$ for an increasing map of the score that puts the three predictors on one scale (here Platt scaling [29] fitted to confidently labelled ClinVar variants, which the fixed-cut baselines of Section 4.1 also use). Because q is strictly increasing, every CertiVar call is the same on q as on s, and no ClinVar label affects any call.

Per-class thresholds. The nonconformity of label 1 is $r_1 = 1 - q$ and of label 0 is $r_0 = q$. From each residue in $\mathcal{A}$ we keep one variant chosen uniformly at random. For class k, let n_k be the number of kept variants with label k and $r^{(k)}_{(1)} \le \cdots \le r^{(k)}_{(n_k)}$ their sorted nonconformities. The threshold is $\hat{t}_k = r^{(k)}_{(j_k)}$ (∞ if $j_k > n_k$), where j_k is the in-expectation rank or the probably approximately correct (PAC) rank at confidence level δ_{P} ($j_k = n_k + 1$ if no j qualifies):

$$j_k = \lceil (n_k + 1)(1 - \alpha_k) \rceil \text{ (in expectation)} \quad \text{or} \quad j_k = \min\{\, j : \; F_{\mathrm{Beta}(j,\, n_k - j + 1)}(1 - \alpha_k) \le \delta_{\mathrm{P}} \,\} \text{ (PAC).} \quad (1)$$

The prediction set is $C(q) = \{k : r_k \le \hat{t}_k\}$; CertiVar calls LoF if $C = \{1\}$, functional if $C = \{0\}$, and abstains otherwise. In effect it places two cut-offs on the score, $\hat{t}_0$ and $1 - \hat{t}_1$, each an order statistic of one class: a variant below both is called functional, one above both is called LoF, and one between them is not called (Figure 1). A functional call on a LoF variant requires $1 \notin C$, so per-class coverage bounds both error rates.

Certified likelihood ratios. The ACMG/AMP guidelines [3] grade evidence in qualitative categories; in their Bayesian adaptation, supporting, moderate, strong and very strong evidence towards pathogenicity correspond to a likelihood ratio above 2.1, 4.3, 18.7 and 350, and supporting, moderate and strong evidence towards benignity to one below 0.48, 0.23 and 0.053 [4, 5]. We reuse these values as a reporting scale, test levels up to strong, and write, for example, *LoF-supporting* for LoF calls whose likelihood ratio exceeds 2.1. Our likelihood ratios are defined with respect to the assay outcome Y and averaged over all variants that receive a call, as OddsPath is for a functional assay [5]; they are not the local, score-specific ratios that ClinGen uses for the computational criteria PP3/BP4 [2]. A level thus describes the call class as a whole, need not hold for each call, and must not be combined with the same assay's functional evidence (PS3/BS3) or with PP3/BP4 from the same predictor. For thresholds t, the call rates are $\lambda_k(t) = \Pr(\text{call LoF} \mid Y = k)$ and $\phi_k(t) = \Pr(\text{call functional} \mid Y = k)$, and $\mathrm{LR}^+ = \lambda_1/\lambda_0$, $\mathrm{LR}^- = \phi_1/\phi_0$.

For each α on a grid G we compute thresholds on $\mathcal{A}$ ($\alpha_0 = \alpha_1 = \alpha$), call one variant per residue of $\mathcal{B}$, and bound the four rates by one-sided Clopper–Pearson bounds [30] at level $\epsilon = \delta/(2|G|)$: for x calls among n variants of a class, the lower bound is $F^{-1}_{\mathrm{Beta}(x,\, n-x+1)}(\epsilon)$ (0 if $x = 0$) and the upper bound $F^{-1}_{\mathrm{Beta}(x+1,\, n-x)}(1-\epsilon)$ (1 if $x = n$, including $n = 0$). Writing $\underline{\lambda}_1, \overline{\lambda}_0, \overline{\phi}_1, \underline{\phi}_0$ for these bounds, the certified bounds are $L^+(\alpha) = \underline{\lambda}_1/\overline{\lambda}_0$ and $U^-(\alpha) = \overline{\phi}_1/\underline{\phi}_0$ ($U^- = \infty$, never meeting a level, if $\underline{\phi}_0 = 0$; $\overline{\lambda}_0 > 0$ always; for point estimates, $x/0 = \infty$ for $x > 0$, and $0/0$ is undefined and never selected). We write $\hat{t}(\alpha)$ for the pair $(\hat{t}_0, \hat{t}_1)$ at α and call the share of the kept variants of $\mathcal{B}$ that receive a call of a given type its *yield*. For each evidence level, Algorithm 1 returns the α with the largest yield among those whose certified bound meets the level. Because $\hat{t}_0$ and $\hat{t}_1$ both decrease weakly in α, no variant can be called LoF at one α and functional at another (Section 3.1). Line 7 requires, for functional calls, α at least as large as the LoF choice of the same strength, so that the two call types of one strength come from an ordered pair of thresholds; this is a reporting convention, not needed for validity, and can only lower functional yields. The *fixed-sequence* variant replaces the Bonferroni level $\delta/(2|G|)$ by $\delta/(2L)$, where L is the number of evidence levels tested for a call type, tests the grid in an order fixed in advance, and stops at the first α that does not meet the level (Theorem 2). We test LoF levels in increasing and functional levels in decreasing α, because few variants are called functional at small α. All thresholds and bounds cost $O(|G|\, n \log n)$ for n residues, about 5 ms per gene on a laptop CPU.

Algorithm 1 CertiVar for one gene and one evidence level (Bonferroni version)

Require: scores q; assay labels split by residue into $\mathcal{A}, \mathcal{B}$; grid G; δ; levels τ (LoF) and τ' (functional)
1: keep one variant per residue of $\mathcal{A}$ and of $\mathcal{B}$
2: **for** $\alpha \in G$ **do**
3: thresholds $\hat{t}_0, \hat{t}_1$ on the kept variants of $\mathcal{A}$ by (1); call the kept variants of $\mathcal{B}$
4: $L^+(\alpha) \leftarrow \underline{\lambda}_1/\overline{\lambda}_0$, $U^-(\alpha) \leftarrow \overline{\phi}_1/\underline{\phi}_0$ (Clopper–Pearson, level $\delta/(2|G|)$ each)
5: **end for**
6: $\hat{\alpha}^+ \leftarrow \arg\max_{\alpha \in G}\{\text{share called LoF on } \mathcal{B} : L^+(\alpha) > \tau\}$
7: $\hat{\alpha}^- \leftarrow \arg\max_{\alpha \in G}\{\text{share called functional on } \mathcal{B} : U^-(\alpha) < \tau',\ \alpha \geq \hat{\alpha}^+\}$ (over all $\alpha \in G$ if $\hat{\alpha}^+$ does not exist)
8: **output** LoF calls at $\hat{\alpha}^+$ and functional calls at $\hat{\alpha}^-$ with their certified levels (no calls of a type whose set is empty); abstain otherwise

Checking class-conditional invariance. The guarantees transfer to a target set only if, within each class, target scores follow the calibration law (Assumption 2 below), which makes the target score law a mixture $\pi F_1 + (1-\pi)F_0$ of the two class laws. We test this necessary condition on one target variant per residue with $D = \min_\pi \sup_x \left|\hat{F}_T(x) - \pi\hat{F}_1(x) - (1-\pi)\hat{F}_0(x)\right|$, where $\hat{F}_T$, $\hat{F}_1$ and $\hat{F}_0$ are the empirical distribution functions of the target scores and of the two classes in $\mathcal{A}$, and π runs over a grid of step 0.005. The null distribution of D is approximated by a bootstrap that redraws both calibration classes and a target sample of the same size from the fitted mixture (300 resamples). After a rejection we do not report calls for that target set; acceptance does not establish Assumption 2.

3.1 Guarantees

Let Q be the law of a variant obtained by drawing a residue from the gene's population of assayed residues and then one of its variants uniformly at random (binary-labelled, non-splice variants outside the ClinVar labelled set). Every rate and likelihood ratio below is taken under Q, so it weights residues equally.

Assumption 1 (Clustered sampling)**.** Residues in $\mathcal{A}$, in $\mathcal{B}$ and of a test variant are independent draws from one residue population; within a residue, the kept or tested variant is chosen uniformly at random; q does not depend on these data.

Assumption 2 (Class-conditional invariance)**.** Target variants have, for $k \in \{0,1\}$, the same law of q given $Y = k$ as a test variant under Assumption 1; their LoF share $\pi = \Pr(Y = 1)$ may differ.

Proposition 1 (Per-class coverage)**.** *Under Assumption 1, for a fixed α_k, with the in-expectation rank and conditionally on n_k, $1 - \alpha_k \leq \Pr(k \in C \mid Y = k, n_k) \leq 1 - \alpha_k + 1/(n_k + 1)$, the upper bound holding when there are no ties.*

Proposition 2 (High-probability per-class coverage)**.** *Under Assumption 1, for a fixed α_k and with the PAC rank, $\Pr(k \in C \mid Y = k, \hat{t}_k) \geq 1 - \alpha_k$ with probability at least $1 - \delta_{\mathrm{P}}$ over the calibration data for each k, and for both classes with probability at least $1 - 2\delta_{\mathrm{P}}$.*

Proposition 3 (Coverage after selection)**.** *Under Assumption 1, for any $\hat{\alpha} \in G$ chosen from data independent of the test variant (for example by Algorithm 1), the in-expectation rank gives $\Pr(k \in C \mid Y = k) \geq 1 - \max G$; the PAC rank at level $\delta_{\mathrm{P}}/|G|$ gives $\Pr(k \in C \mid Y = k, \hat{t}_k) \geq 1 - \hat{\alpha}$ for each k with probability at least $1 - \delta_{\mathrm{P}}$.*

Theorem 1 (Certified likelihood ratios)**.** *Under Assumption 1, conditionally on $\mathcal{A}$ and with probability at least $1 - \delta$ over $\mathcal{B}$, $\mathrm{LR}^+(\hat{t}(\alpha)) \geq L^+(\alpha)$ for every $\alpha \in G$ simultaneously; the same holds for $\mathrm{LR}^- \leq U^-$ with a second δ. Because the event does not depend on the level, each call type returned by Algorithm 1 reaches its certified level at every evidence level reported, with probability at least $1 - \delta$ (both call types with probability at least $1 - 2\delta$), whichever α is selected.*

Theorem 2 (Fixed-sequence certification)**.** *Under Assumption 1, fix a level τ, a confidence δ' and an order of G chosen in advance. Reject $H_\alpha : \mathrm{LR}^+(\hat{t}(\alpha)) \leq \tau$ when $\underline{\lambda}_1/\overline{\lambda}_0 > \tau$ with both endpoints at level $\delta'/2$, stop at the first α not rejected, and select the rejected α with the largest yield; the rejected set is a prefix of the*

Table 1: Datasets. *ClinVar labelled*: missense variants with ≥2 review stars and a P/LP or B/LB call (used for the common score scale and an exploratory check). *Assayed*: missense variants with a binary assay label outside that set; *assayed VUS*: the subset classified VUS. LoF shares are variant-level. Labels: authors' classes or a rule fitted to the assay's controls (Section 4.1).

			ClinVar labelled		Assayed			Assayed VUS	
Gene	Assay	Labels	n	P/LP	n	Residues	LoF	n	LoF
BRCA1	SGE [31]	authors	345	156	1,617	307	0.161	272	0.173
BRCA2	SGE [32]	controls	309	94	3,435	676	0.130	704	0.207
BAP1	SGE [33]	authors	14	6	4,294	696	0.213	1,069	0.166
RAD51C	SGE [34]	authors	13	10	2,225	355	0.259	550	0.396
MSH2	DMS [35]	authors	178	121	2,906	893	0.060	1,542	0.067
Total			859	387	14,477	2,927	0.164	4,137	0.167

order. (a) Conditionally on $\mathcal{A}$, the selected LoF calls have $\mathrm{LR}^+ > \tau$ with probability at least $1-\delta'$. (b) If each of L levels is tested in this way with $\delta' = \delta/L$, every level claimed for LoF calls holds simultaneously with probability at least $1-\delta$. The same holds for functional calls with $\bar{\phi}_1/\underline{\phi}_0 < \tau'$, and for both call types together with probability at least $1-2\delta$.

Corollary 1 (No prior needed)**.** *Under Assumptions 1 and 2, for target variants drawn residue first as under Q, Propositions 1–3 and Theorems 1–2 hold whatever their LoF share $\pi \in (0,1)$: for a fixed α, LoF variants are called functional at rate at most α_1, and the certified likelihood ratios with respect to the assay outcome need no prior. For any $\alpha^+, \alpha^- \in G$, the LoF calls at α^+ and the functional calls at α^- are disjoint.*

Propositions 1–2 are the class-conditional and training-conditional conformal guarantees [12, 13, 15, 16] under Q; Theorems 1–2 apply the union bound of risk control [20, 21] and the fixed-sequence argument of Learn-then-Test [22] to a *ratio* of class-conditional rates of an abstaining rule. Monotonicity of both ranks in α makes coverage at a selected α at least that at $\max G$ and keeps the call types disjoint. Proofs are in Section S1.

Scope. After selection, per-class coverage is guaranteed only at $1 - \max G$; Theorem 2 certifies levels, not bounds. Assumption 1 treats residues as independent draws (our random splits draw without replacement; the oracle design does not), and a contiguous region need not be exchangeable with the rest; Assumption 2 holds only approximately for VUS (Section 4.3) and not for variants whose assay result would be intermediate.

4 Experiments

4.1 Data and protocol

We took every GRCh38 missense single-nucleotide variant in BRCA1, BRCA2, BAP1, RAD51C and MSH2 that is in ClinVar [1] (release of 26 October 2026) or was assayed, annotated each on its MANE Select transcript with the Ensembl Variant Effect Predictor [36] (AlphaMissense [37], REVEL [38]) and computed ESM-2 650M [39] log-likelihood ratios by wild-type marginals [40]. Labels come from five MaveDB [41] score sets (Table 1; accessions in Section S5): the authors' classes for BRCA1, BAP1 and RAD51C and the authors' threshold for MSH2 [35]. The BRCA2 score set has no deposited classes, so we fitted Gaussians to its synonymous and nonsense variants (LoF if the posterior is at least 0.95, functional if at most 0.05); applied where the authors' classes exist, this rule disagreed with them on 216 of 4,023 BAP1 and none of 1,762 BRCA1 variants (Section 4.6 gives other label rules). We excluded 729 splice-region and 1,274 intermediate variants. One Platt map per predictor used the 859 ClinVar missense variants with at least two review stars and a pathogenic or likely pathogenic (P/LP) or benign or likely benign (B/LB) call (45.1% P/LP), which are excluded from every evaluation set except the exploratory ClinVar check, leaving 14,477 assayed variants at 2,927 residues. AlphaMissense has an AUROC of 0.946 (95% residue-bootstrap interval

Table 2: Per-class coverage at $\alpha = 0.1$ (residue-weighted; 50 random halvings per gene). Last column: worst pair when the calibration half is subsampled to a LoF share of 0.05 (100 halvings); fixed cut-offs use no calibration data.

		Worst-class coverage				Worst pair,
Method	Pairs	mean	worst pair	pairs < 0.85	Abstention	share 0.05
Per-class (ours)	15	0.901	0.896	0	0.35	0.899
Pooled conformal	15	0.836	0.545	5	0.42	0.113
Label-shift weighted, true share	15	0.836	0.559	5	0.42	0.598
Label-shift weighted, EM share	15	0.722	0.039	9	0.36	0.038
Fixed tiers ($q \geq 0.9$, $q \leq 0.1$)	15	0.871	0.663	5	0.53	–
AlphaMissense classes	5	0.725	0.540	4	0.11	–
REVEL PP3/BP4 thresholds	5	0.700	0.445	5	0.42	–

0.922–0.965) on the ClinVar set, 0.840 (0.829–0.852) on the other assayed variants and 0.779 (0.757–0.799) on assayed VUS. Baselines are pooled (marginal) split conformal prediction, label-shift-weighted conformal prediction [28] with the true target share or its expectation–maximization (EM) estimate [26], fixed tiers ($q \geq 0.9$ LoF, $q \leq 0.1$ functional), the published AlphaMissense classes (> 0.564, < 0.34) and the REVEL PP3/BP4 thresholds (≥ 0.644, ≤ 0.290) [2]; the published cut-offs target pathogenicity, so we judge them only as assay classifiers. Coverage is residue-weighted as in Q; *worst-class coverage* is the smaller of the mean LoF and mean functional coverages over repetitions. Every split is by residue (designs in Table S7).

4.2 Per-class coverage and a shift in the calibration LoF share

Over 50 random halvings of each gene's residues, the in-expectation rule at $\alpha = 0.1$ gave worst-class coverage between 0.896 and 0.908 in all 15 gene–predictor pairs, none more than one Monte Carlo standard error (0.002–0.010) below the target (Table 2; per pair in Table S8). Pooled conformal fell to 0.545 on LoF variants and below 0.85 in 5 pairs, because a marginal threshold controls coverage averaged over the two classes, not within each; the published cut-offs covered functional variants as little as 0.540 (AlphaMissense, BAP1) and 0.445 (REVEL, MSH2). The per-class rule abstained on 35% of variants on average (17–47%), against 42% for pooled conformal and 11% for the AlphaMissense classes. Calibrating on all variants instead of one per residue lost coverage through its variant weighting (Section S7).

To compare the methods under a prevalence shift, we subsampled each calibration half to a LoF share of 0.05 to 0.5 and tested at the natural share (100 repetitions; Figure S5A). Subsampling by class leaves each class's score law unchanged, so Corollary 1 predicts the per-class result: worst-class coverage stayed between 0.897 and 0.925 in every pair at every share, with abstention rising from 37% to 83% as LoF variants became scarce. Pooled conformal calibrated at share 0.05 covered LoF variants at 0.113 in its worst pair, and label-shift-weighted conformal fell to 0.53–0.60 with the true target share and to 0.04 with the EM estimate, which missed the LoF share by 0.255 on average because ClinVar pathogenicity and assay loss of function differ in definition as well as in class share.

4.3 Reliability across splits and a real change of population

With 17 to 93 LoF residues per calibration half, a single split can miss the in-expectation guarantee: LoF coverage on a test half was below 0.90 in 36.4% of 200 splits per pair and functional coverage in 43.6% (Figure S4B). PAC thresholds ($\delta_P = 0.1$) reduced these to 7.9% and 10.8% at the price of abstaining on 58.4% of variants against 35.4%; computed exactly on all of a gene's residues, which removes the test half's sampling noise, PAC coverage fell below 0.90 in 4.8% (LoF) and 7.1% (functional) of draws and in at most 10.0% in any pair, within δ_P. Calibrated on non-VUS variants of one half of each gene's residues and applied to the assayed VUS of the other half (residue-weighted LoF share 17.0% in calibration, 19.8% in VUS), in-expectation thresholds kept LoF coverage between 0.854 and 0.958 and functional coverage between 0.841 and 0.942, each below 0.90 in 7 of 15 pairs (Figure S4C); PAC thresholds abstained on 81.7% of VUS and still fell below 0.90 for functional variants in 2 pairs (lowest 0.881). On VUS, Assumption 2 holds only approximately.

Table 3: Selection rules against the population likelihood ratio (oracle design, G_4, $\delta = 0.05$, the same 1,500 draws). Guarantee: finite-sample validity after selection for all levels at once. Bounds and levels held: draws in which every claimed bound (fixed sequence certifies levels only; a point estimate is no bound) or level held (%). Last four columns: draws certifying each level (%).

	Finite-sample	Bounds	Levels	LoF calls		Functional calls	
Selection rule	guarantee	held	held	supp.	mod.	supp.	mod.
Certified (ours), Bonferroni	yes, all levels	99.7	99.9	86.7	13.3	44.9	1.5
Certified (ours), fixed sequence	yes, all levels	–	99.9	74.9	14.4	43.8	1.5
Clopper–Pearson, uncorrected	no	97.9	99.5	95.9	35.1	68.0	6.9
Katz interval, Bonferroni	no	98.0	99.7	99.3	52.5	52.4	0.2
Percentile bootstrap	no	57.6	72.5	99.9	78.1	90.0	45.1
Percentile bootstrap, +0.5 counts	no	75.4	88.2	99.9	76.6	89.3	37.3
Point estimate	no	–	42.1	100.0	98.2	99.8	75.7

Table 4: Mixture-consistency check ($p < 0.05$; one target variant per residue). Coverage: mean over target sets of the smaller per-class coverage in each set (a per-set measure, unlike Table 2) of in-expectation thresholds ($\alpha = 0.1$) set on the same calibration data.

		Target	Rejected	Worst-class coverage	
Target set	Assumption 2	sets	(%)	accepted	rejected
Random half of the residues	holds by design	1,500	4.7	0.880	0.880
Assayed VUS at other residues	approximately	750	8.5	0.845	0.816
Held-out contiguous quarter	not assumed	1,500	43.9	0.837	0.788
Another gene's calibration data	not assumed	300	77.3	0.745	0.644
Unassayed VUS, other residues	unknown (no labels)	120	100.0	–	–

4.4 Certified likelihood ratios

Treating each gene's assayed residues as the population, we drew calibration (30%) and certification (50%) residues with replacement, 100 times per pair, ran Algorithm 1 with $\delta = 0.05$ on the four-point grid $G_4 = \{0.05, 0.10, 0.20, 0.30\}$, and computed the population likelihood ratio of every selection exactly, applying all rules to the same draws (Table 3); Assumption 1 holds here by construction, and Section 4.6 tests held-out regions. Every bound claimed in a draw held in 99.7% of the 1,500 draws for the certified rule (Theorem 1 guarantees at least 90%). Percentile-bootstrap bounds all held in only 57.6% of draws, 65.3% with a Bonferroni correction and 75.4% with 0.5 added to every count; 59% of their failures came from certification samples with no false call of that type. Point-estimate claims all held in only 42.1% of draws (Table 3 and Figure S3). Uncorrected Clopper–Pearson bounds and Katz intervals with a Bonferroni correction, which have no guarantee after selection, held in 97.9% and 98.0% of draws and certified more often: Katz bounds reached LoF-moderate in 52.5% of draws against 13.3%.

CertiVar certified LoF-supporting calls in 86.7% of draws (calling 25.2% of variants), functional-supporting calls in 44.9% (61.4% called), LoF-moderate in 13.3% and functional-moderate in 1.5%, and never a strong level. Per-class coverage at the selected α averaged 0.81 (LoF) and 0.80 (functional), above the guaranteed 0.70 (Proposition 3). Fixed-sequence testing with the level-simultaneous guarantee of Theorem 2(b) traded supporting for moderate evidence: on G_4 it certified LoF-moderate in 14.4% of draws (Bonferroni: 13.3%) but LoF-supporting in 74.9% (86.7%), and on a 20-point grid of values from 0.02 to 0.40, LoF-moderate in 12.7% (6.4%) but LoF-supporting in 50.3% (74.8%; Figure S5B). On held-out fifths of the residues, empirical likelihood ratios respected the certified bounds in 98.7% of 2,091 cases (bootstrap: 76.8%); on VUS at other residues, LoF-supporting held in all 443 certified claims and LoF-moderate in 25 of 28. An exploratory check against ClinVar labels is in Section S7.

4.5 Checking invariance, and where CertiVar applies

The mixture-consistency check rejected 4.7% of random halves (3.0–7.7% by gene), where Assumption 2 holds by design (Table 4). It rejected 43.9% of held-out contiguous quarters, 77.3% of transfers from another gene and every set of unassayed ClinVar VUS at residues the BRCA1 and BRCA2 assays did not reach, where the thresholds abstained on only 8.9% of variants against 35% on assayed residues. Rejected targets had lower coverage than accepted ones, but the check has little power for assayed VUS (8.5% rejected), so passing it is not evidence that the guarantees transfer. Of the 147 ClinVar VUS that were unassayed but lay at assayed residues, 105 are in MSH2; there the check rejected 40% of samples with AlphaMissense and 35% with REVEL but none with ESM-2, whose thresholds gave certified LoF-supporting calls to 19.1 and functional-supporting calls to 5.0 of the 105 on average over 20 splits (Tables S5 and S6). The 4,010 unassayed BRCA1 and BRCA2 VUS at other residues are not called, because the check rejected every sample of them.

4.6 How much data, from where, and with which labels

Calibration data had to come from the same gene: calibrating on the other four genes left 8 of 15 pairs below 0.85 (worst 0.487), consistent with gene-specific evidence strength [10, 11]. Calibrating on three contiguous quarters of a protein and testing on the fourth gave worst-class coverage of 0.860–0.923, and 88.8% of 1,891 certified bounds held against the held-out quarter's empirical likelihood ratios, below the 95% that Theorem 1 guarantees under Assumption 1. Intermediate variants, which no guarantee covers, were called LoF in 19.6% and functional in 36.6% of cases. Relabelling BRCA2, BAP1 and RAD51C with alternative rules (Table S3) changed yields but not validity: the lowest per-class coverage over predictors stayed between 0.894 and 0.902, and certified bounds held in 2,530 of 2,533 oracle selections. Calibration size and class-boundary strata are in Sections S3 and S4.

5 Discussion

Thresholds selected on point estimates or percentile-bootstrap bounds often claimed more evidence than held. These are call-averaged ratios, not the local ratios of ClinGen calibrations, so this does not show that published PP3/BP4 thresholds are wrong, only that point estimates and bootstrap bounds can overstate evidence at single-gene sample sizes. Confidence bounds for the ratio, exact or Katz, held in at least 97.9% of draws; the exact certificate is the conservative choice that keeps a guarantee at every sample size. CertiVar applies to genes with a functional assay: to audit whether a predictor's claimed evidence strength holds against the assay, and to call the variants the assay missed at assayed residues, for predictors whose scores pass the mixture check. The cost is conservatism: certified evidence was mostly supporting-level, and PAC thresholds abstained on 58% to 82% of variants.

Limitations. The labels are assay classes, not clinical pathogenicity [5], and the ACMG/AMP levels are reused as a reporting scale; clinical use requires an assay validated against clinical classifications, as its authors did for BRCA1 [31] and BRCA2 [32]. Our BRCA2 labels come from the assay's controls (Table S3). A certified level describes the calls as a class and weights residues equally, so a borderline call can carry less evidence than its class. The guarantees do not cover intermediate, splice-region or out-of-region variants; intermediate variants are a hazard, because whether a VUS would score intermediate is unknown before it is assayed, and 19.6% of them were called LoF. The mixture check tests only a necessary condition and missed most VUS shifts. Results cover five cancer-predisposition genes. No predictor was trained on these assay labels, but REVEL was trained on labelled variants and the AlphaMissense classes were set on ClinVar, so both may overlap our evaluation sets [38, 42]; ESM-2, trained without variant labels, gave the same coverage and validity.

Conclusion. CertiVar turns a variant-effect predictor and a few hundred assayed residues of a gene into LoF, functional or no calls with finite-sample guarantees on per-class error rates and on call-averaged likelihood ratios for the assay after threshold selection, without an estimate of how common damaging

variants are, and checks whether a target set can inherit them. Code, data and results (release v1.1.1): `https://github.com/aaryavvatts-lab/certivar`.

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